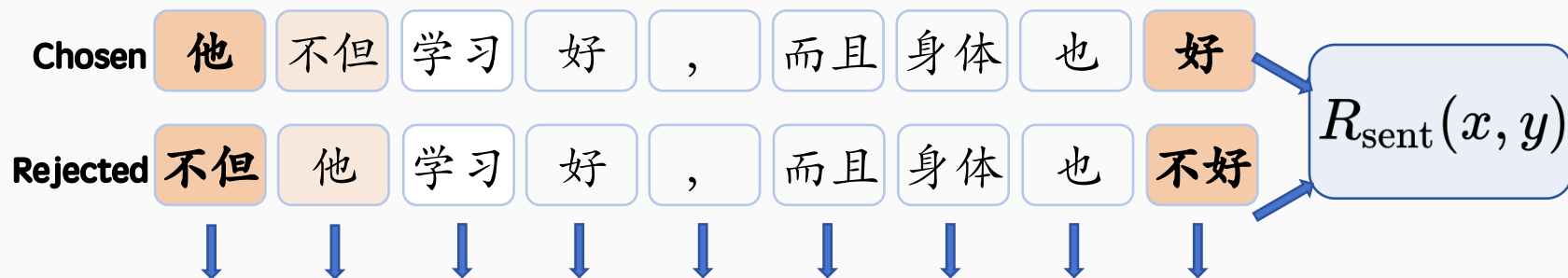


EARM Training

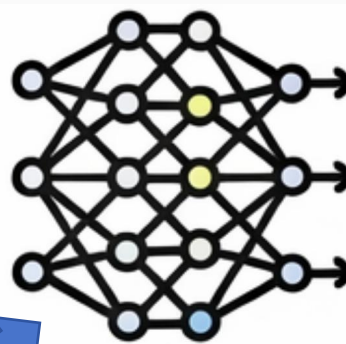


$$R_{\text{weighted}}(x, y) = \frac{\sum_{i \in \text{resp}} w_i \cdot s_i}{\sum_{i \in \text{resp}} w_i}$$

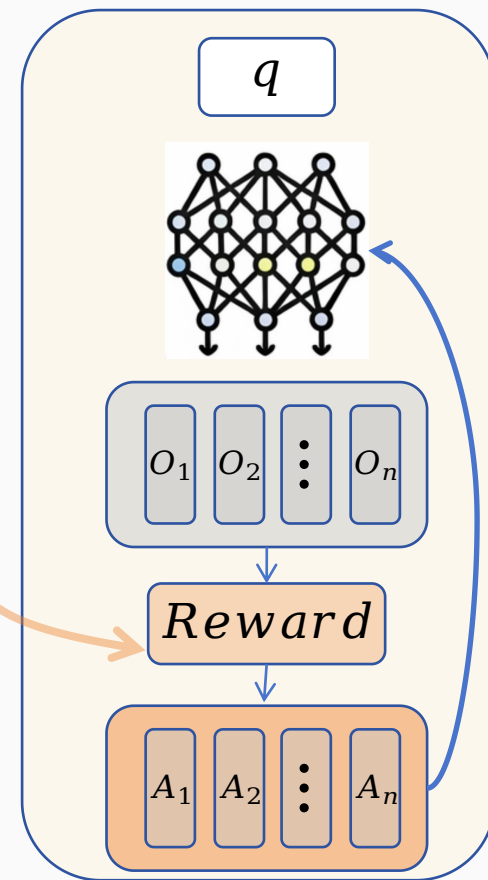
Token Loss : $\mathcal{L}_{\text{weighted}}$

Sentence Loss : $\mathcal{L}_{\text{rank}}, \mathcal{L}_{\text{margin}}$

$$\mathcal{L}_{\text{EARM}} = \mathcal{L}_{\text{rank}} + \lambda \mathcal{L}_{\text{weighted}} + \mu \mathcal{L}_{\text{margin}}$$



GRPO Training



pivot tokens continuation tokens unchanged tokens